WalletFlow — Progress Summary

COT 5480 Semester Project · Team WalletFlow: James Clifford, Cody Gilstrap, Aden Brady · November 2026

What works

- **Add and store:** all seven fields are validated (ID, amount in integer cents, type, urgent flag, priority 1–3, category, ISO 8601 arrival time). Duplicate IDs and invalid fields are rejected with a clear message, and a rejected add leaves stored data unchanged.
- **Find by ID:** returns the full record, or *None* for an unknown ID.
- **Ordinary processing (FIFO)** and **urgent processing**: urgent payments go first, highest priority first, with equal priorities in arrival order. Processing sets status to *processed*, removes the payment from pending work, and keeps the record for search and reporting. An empty queue returns a “no pending transactions” result.
- **Testing:** 13 *unittest* tests (python3 -m unittest -v) print input, expected, and actual results; all pass. A 10-transaction sample set and *demo.py* show every implemented operation. An extra randomized check of 10,000 transactions matched a reference sort.

Current design choices

Operation	Structure	Reason
Add, duplicate check, find	dict keyed by txn_id	Average constant-time insert, membership test, and lookup
Ordinary processing	collections.deque	append / popleft give constant-time first-in, first-out
Urgent processing	heapq of (–priority, seq, txn)	Highest priority pops first; arrival counter seq breaks ties in arrival order
Records vs. pending work	Separate store and queues	Processing removes work from a queue but never deletes the record

Changes from the proposal

- Added an internal arrival counter (*seq*) so ties never depend on identical timestamps.
- Added a single *WalletFlow* class that owns the store and the queues, so adding a transaction also queues it.
- Added an *other* category. Delivered as Python files plus a demo script; a notebook will be added for the final demo.
- Urgent processing is finished early, ahead of the original Weeks 12–14 plan.

What remains

- **Fixes found in code review:** consistent time-zone handling (required before sorting by date), returning copies from find so stored records cannot be changed from outside, trimming spaces in IDs on lookup, and a stronger repeat-processing test.
- **Features:** sorted report (arrival time, then ID) and total processed amount by category.
- **Final deliverables:** remaining edge-case tests, list-scan vs. dictionary lookup comparison at 3+ sizes, complexity and space analysis, report, README, and video.

Updated completion plan

Week	Tasks	Owner
12 (Nov 9–15)	Review fixes; sorted report; category totals	James (fixes, validation), Aden (report, totals), Cody (processing review)
13 (Nov 16–22)	Edge-case tests per module; data generator; lookup benchmark; integration test. **Code complete Nov 22.**	Each owner for own tests; Aden (generator); Cody (benchmark); James (integration)
14 (Nov 23–29)	Complexity and space table; benchmark results and interpretation; report draft; cross-review	James and Cody (analysis); Aden (report); all review
15 (Nov 30–due)	Clean-run check of notebook and README; record video; submit	All

AI use statement

Anthropic Claude was used to draft the project plan, generate the initial implementation and tests, review the code, and draft this summary. The team reviewed, ran, and tested the code and remains responsible for the design decisions, implementation, testing, and final explanations. OpenAI ChatGPT was used for the proposal, as previously stated.